Artemis: Safe and Fuel Efficient Decision Making in an Uncertain Lunar Mission

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Abstract

We study a small but structured reinforcement learning problem inspired by Artemis II style lunar mission planning, where an agent must trade fuel efficiency against safety under hidden stage dependent risks. The environment is a finite horizon Markov decision process with 48 non terminal states, three actions (safe, risky, fallback), stochastic transitions, and a hard fuel constraint. We compare four tabular methods against an oracle computed by value iteration on the true MDP: Q learning, UCB Q, model based value iteration with a learned model, and posterior sampling for reinforcement learning. Across a 2,000 episode, five seed benchmark, all methods learn meaningful policies, but UCB achieves the best overall balance of success rate, return, and regret, while model based planning becomes competitive once enough data has been collected. PSRL obtains the lowest hazard rate, but shows a safety and performance trade off rather than dominance on success. The report documents the exact environment, protocol, and caveats needed to reproduce and interpret the results[1].

1. Introduction

Sequential decision making under uncertainty is central to reinforcement learning, especially when exploration itself can be costly (Sutton & Barto, 2018). In safety critical domains, a learner must discover useful behaviors without repeatedly taking catastrophic actions. This tension is a natural fit for a simplified mission planning problem: some actions preserve resources but expose the agent to hidden risk, while safer actions consume resources more aggressively and can make long term success impossible.

Motivated by Artemis II style lunar mission planning, we study a compact episodic MDP in which a spacecraft advances through mission stages while managing fuel, hazard exposure, and a one shot fallback maneuver. Episodes have a finite horizon, but they may terminate earlier if the agent reaches mission success or failure. The project question is:

> Can uncertainty aware exploration methods learn safer and more fuel efficient strategies than standard Q learning in a simplified lunar mission with hidden risks?

The environment is deliberately small enough for exact planning, which lets us compute an oracle policy by value iteration and use it as a principled upper bound. This makes the project useful both as an application study and as a controlled benchmark for comparing exploration strategies. The work therefore connects classic tabular RL ideas such as Q learning (Watkins & Dayan, 1992), optimism based exploration (Auer et al., 2002), model based planning (Puterman, 1994; Sutton & Barto, 2018), and posterior sampling (Osband et al., 2013).

Our main contributions are:

- a fully specified finite horizon lunar mission MDP with stage dependent hidden risk,
- a comparison of four tabular RL methods under a common protocol,
- an oracle baseline and regret computation relative to the true environment,
- robustness experiments under harder mission variants,
- a discussion of what the learned policies reveal about the fuel and risk trade off.

2. Environment and MDP Formulation

2.1. Scenario

A mission consists of four decision stages followed by terminal completion: Launch, Earth departure, midcourse transfer, lunar approach, and mission completion. At each non terminal stage, the agent chooses one of three actions: *safe route*, *fuel efficient risky route*, or *fallback maneuver*. Safe and fallback actions consume fuel but have no direct catastrophic branch. The risky action can preserve fuel on its

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[1]for reproducibility, one can find the source code here

Table 1. Core environment specification.

Component	Specification
Stages	1 = Launch, 2 = Earth departure, 3 = Mid-course transfer, 4 = Lunar approach, 5 = mission completion
State	(k, f, h, b) with fuel $f \in \{0, 1, 2\}$, hazard $h \in \{0, 1\}$, fallback used $b \in \{0, 1\}$
Actions	Safe, risky, fallback
Initial state	$(1, 2, 0, 0)$
Terminal success	Reach stage 5 with fuel > 0
Terminal failure	Catastrophic transition or fuel dropping below 0 before stage 5
Reward	$+10$ stage advance, $+50$ final success bonus, -5 entering hazard, -10 using fallback, -100 failure

safe branch but may enter a hazardous state or fail catastrophically.

The core decision tension is simple: preserving fuel early is strategically valuable because the final stage requires arriving with fuel remaining, but the actions that preserve fuel are exactly those whose transition probabilities are hidden and risky.

2.2. State, actions, and horizon

We model the mission as a finite horizon MDP

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, H)$$

with horizon $H = 4$ decision steps. Each non terminal state is

$$s = (k, f, h, b),$$

where:

- $k \in \{1, 2, 3, 4\}$ is the current mission stage,
- $f \in \{0, 1, 2\}$ is the remaining fuel level,
- $h \in \{0, 1\}$ indicates normal or hazardous status,
- $b \in \{0, 1\}$ indicates whether fallback has already been used.

This yields $4 \times 3 \times 2 \times 2 = 48$ non terminal states, plus absorbing success and failure states. Fuel is encoded numerically as high $= 2$, medium $= 1$, low $= 0$.

2.3. Transition dynamics

Safe route. At stages 1 to 4, the safe action always advances to the next stage and consumes one fuel unit. The hazard state remains normal with probability 0.95 and becomes hazardous with probability 0.05. There is no direct catastrophic failure branch.

Table 2. Risky route transition probabilities under normal hazard status.

Stage	Safe progress	Hazard progress	Failure
1	0.75	0.20	0.05
2	0.70	0.20	0.10
3	0.65	0.20	0.15
4	0.55	0.25	0.20

Fallback maneuver. Fallback can be used once per episode. It always advances to the next stage, consumes one fuel unit, and sets the fallback used flag to 1. It resets hazard to normal with probability 0.90 and otherwise preserves the current hazardous status. Again, there is no direct catastrophic failure branch.

Risky route. The risky action has stage dependent success, hazard, and failure probabilities. On its safe branch, fuel is preserved; on its hazard branch, one fuel unit is consumed. Under normal hazard status, the probabilities are given in Table 2. If the current state is already hazardous, safe progress probability decreases by 0.10 and failure probability increases by 0.10, while hazard progress probability remains unchanged.

Fuel dynamics are deterministic conditional on the transition branch: safe and fallback actions always consume one unit, risky safe progress consumes none, and risky hazard progress consumes one. If an action would reduce fuel below zero, the episode terminates in failure.

2.4. Reward and optimality criterion

The reward function is shaped to make learning easier while preserving the task objective:

$$R(s, a, s') = \begin{cases} +10 & \text{for stage advance} \\ +50 & \text{additional bonus if } s' = s_{\text{succ}} \\ -5 & \text{if } h = 0 \text{ and } h' = 1 \\ -10 & \text{if } a = a_{\text{fallback}} \\ -100 & \text{if } s' = s_{\text{fail}}. \end{cases}$$

Thus, a successful final transition receives a total reward of $+60$. The oracle policy is computed by value iteration on the true MDP using discount factor $\gamma = 0.99$. For evaluation plots and tables, we also report *undiscounted* episodic return, success rate, hazard rate, and cumulative regret relative to the oracle's Monte Carlo expected return. The distinction matters: discounted $V^*(s_0)$ is the planner's internal optimality criterion, while undiscounted episodic return is the human readable score used in the benchmark.

3. Methods and Experimental Protocol

3.1. Algorithms

We compare four tabular agents and one oracle planner.

Q learning. The baseline agent uses the standard update (Watkins & Dayan, 1992; Sutton & Barto, 2018)

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\Big]$$

with ϵ greedy exploration.

UCB Q. UCB Q augments action values with a count based optimism bonus. At state s, the implementation scores legal actions by

$$\text{score}(s,a) = Q(s,a) + c\sqrt{\left(\frac{\log(t)}{N(s,a)+1}\right)}$$

where t is the global action selection timestep, $N(s,a)$ is the number of times action a has been selected in state s, and $c = 5$ controls the strength of the exploration bonus. The $+1$ in the denominator avoids division by zero and plays the role of a pseudocount, although the actual counts start at zero. Learning still uses the Q learning temporal difference update; UCB only changes the action selection rule. Illegal actions are masked before action selection and before planning backups, so agents do not select unavailable actions when fallback has already been used or when an action is not valid in a terminal state.

Model based value iteration. This agent estimates transition probabilities and expected rewards from interaction counts, then periodically recomputes a policy by value iteration on the learned model. In the reported experiments, replanning is done every 5 episodes rather than at every episode.

Posterior sampling for reinforcement learning. The Thompson style agent maintains Dirichlet posteriors over transition distributions and samples a complete MDP at the start of each episode. It then acts greedily with respect to the sampled model for the whole episode.

Oracle. The oracle is not a learning agent. It runs value iteration directly on the true transition and reward model and provides the maximum achievable performance ceiling for this environment. Oracle Monte Carlo evaluation uses greedy rollouts of the oracle policy, not additional exploration noise.

3.2. Experimental design

The evaluation protocol has three layers.

Table 3. Training protocol and fixed hyperparameters used for the reported experiments.

Item	Value
Discount factor γ	0.99
Q learning learning rate α	0.15
Q learning ϵ	0.15, fixed
UCB bonus coefficient c	5
UCB count convention	counts start at 0, denominator uses $N(s,a)+1$
Model based planning frequency	every 5 episodes
PSRL Dirichlet prior	all ones
Single run sanity check	600 episodes, 1 seed
Main benchmark	2,000 episodes, 5 fixed seeds
Trajectory demo	greedy rollout after 2,000 training episodes
Oracle evaluation	10,000 greedy Monte Carlo rollouts of π^*

First, we run a single seed 600 episode training curve to confirm that all agents improve from scratch and to illustrate qualitative differences in learning speed.

Second, we run the main benchmark: five seeds, four agents, and 2,000 episodes per run. Curves are reported as mean $\pm$ one standard deviation across seeds. The final table summarizes the last 100 episodes of each run.

Third, we test robustness under four variants already implemented in the project:

- `harsh_hazard`: hazard penalty changed from -5 to -10,
- `low_fuel`: initial fuel reduced from 2 to 1,
- `risky_x2`: risky route failure probability doubled,
- `weak_fallback`: fallback recovery probability reduced from 0.90 to 0.50.

The main metrics are:

- **success rate**: fraction of successful episodes,
- **return**: undiscounted episodic reward,
- **hazard rate**: average frequency of hazardous transitions per episode,
- **cumulative regret**: $\sum_{i=1}^{n}(J^* - G_i)$, where J^* is the oracle's undiscounted mean return estimated by 10,000 greedy rollouts and G_i is the observed return in episode i.

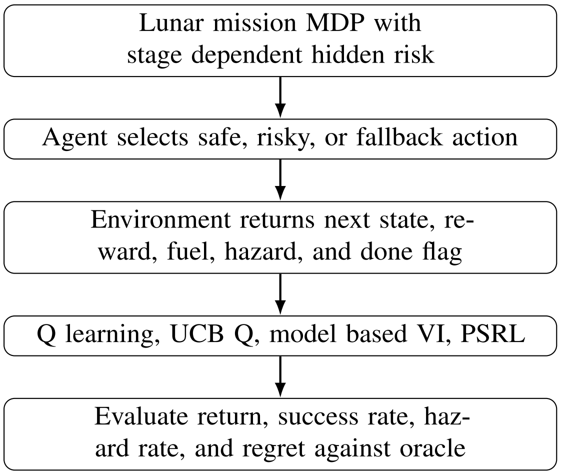


Figure 1. Project workflow used in the implementation and evaluation.

Hyperparameters were fixed for the reported experiments and reused across all variants. We report mean and standard deviation over five seeds, but we do not claim statistical significance beyond that.

4. Results

4.1. Oracle upper bound and structure of the problem

Value iteration on the true model yields a discounted oracle value of $V^*(s_0) \approx -1.535$ at the initial state. A 10,000 episode Monte Carlo evaluation of greedy oracle rollouts gives an undiscounted mean return of approximately -1.1 and a success rate of 47.8%. This is an important caveat: the environment itself is inherently stochastic and difficult, so a success rate in the low 30s should be interpreted relative to that ceiling, not as an absolute failure.

The value structure of the problem explains why the learned policies behave as they do. The largest value difference in the reported oracle analysis is the gap between fuel levels 2 and 1 in early stages. Intuitively, losing one unit of fuel too early removes the possibility of absorbing later stochastic damage. This makes risky actions appealing in early stages because their safe branch preserves fuel, while the safe action always spends one unit. By contrast, when stage 4 is reached with enough fuel, the oracle prefers the safe action to eliminate catastrophic risk.

4.2. Single run learning and episode level behavior

Figure 2 shows a 600 episode single seed run. All agents improve from the beginning. Q learning and UCB rise fastest because they update action values directly from observed transitions. Model based value iteration starts more slowly, since its learned transition model must become reliable before replanning becomes useful. PSRL shows higher early

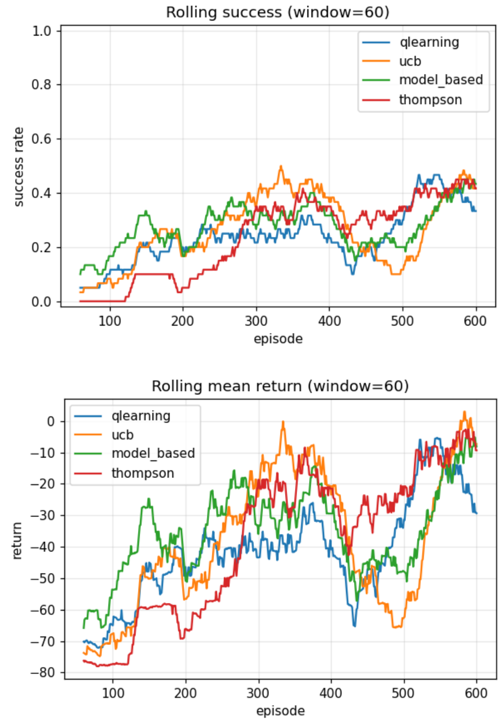


Figure 2. Single run learning curves over 600 training episodes

variance because it explores via sampled MDPs whose posterior uncertainty is initially large.

To complement these averages, we also inspect one greedy rollout after 2,000 training episodes for each agent. These trajectories are used only for intuition and should not be treated as aggregate evidence on their own. They suggest a recurring failure pattern for UCB and model based VI in some seeds: both may invoke fallback in non hazard states, spending fuel without benefit and then reaching the final stage with insufficient margin. By contrast, Q learning's illustrative trajectory most closely matched the oracle pattern of risky early, safe late. This is consistent with the policy level results discussed later, but because these are single trajectories, the main conclusions should still be drawn from the multi seed statistics.

4.3. Multi seed benchmark

Figure 3 illustrates the results coming from the main benchmark. All agents converge well before episode 2,000, but their behaviors differ in meaningful ways.

UCB achieves the fastest and most stable improvement. It also accumulates the lowest cumulative regret, indicating that its count based optimism directs exploration toward useful state action pairs more efficiently than the other strategies. Model based value iteration catches up after an initial data collection phase and becomes competitive

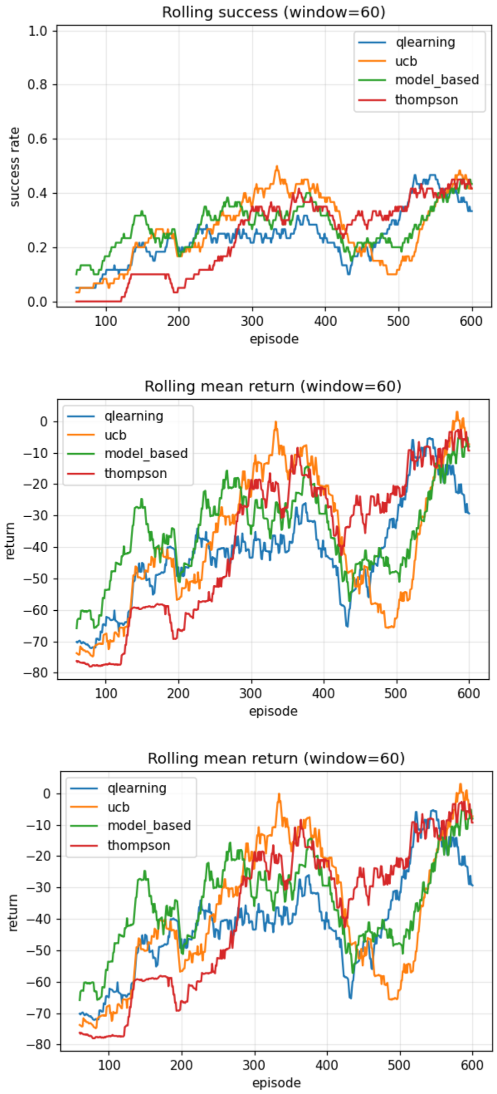


Figure 3. Main benchmark over 2,000 episodes and 5 seeds

once its learned model is sufficiently accurate. Q learning is a strong baseline but explores less efficiently than UCB. PSRL converges more cautiously and exhibits the widest early variance; its final hazard rate is low, but its success rate and regret are worse than UCB and slightly worse than Q learning.

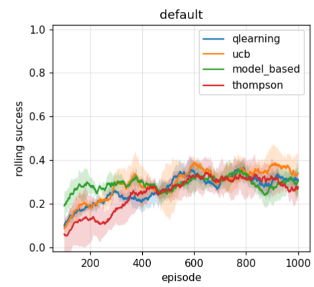
Table 4. Final performance summary over the last 100 episodes of the 2,000 episode, 5 seed benchmark.

Agent	Success	% oracle	Return	Hazard	Regret
UCB	0.344	72%	-22.9	0.127	48,853
Model based VI	0.326	68%	-27.1	0.127	58,935
Q learning	0.318	66%	-27.7	0.122	63,060
PSRL	0.284	59%	-32.2	0.106	71,485

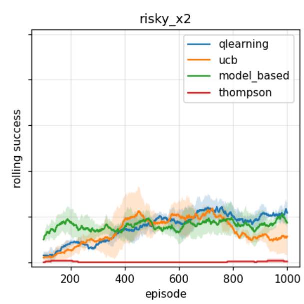
Table 4 makes the ranking explicit. UCB is the strongest overall method on this benchmark. Model based value iteration is a close second once its learned model stabilizes. Q learning is competitive but consistently behind UCB. PSRL is the most cautious: it achieves the lowest hazard rate, but at the cost of lower final success and the highest cumulative regret.

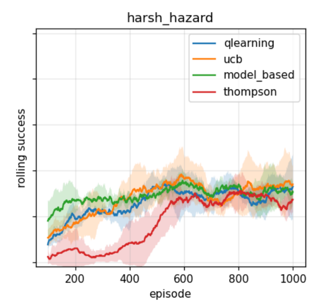
4.4. Robustness to environment variants

The robustness study in Figure 4 shows that the relative ranking is fairly stable but the environment variants reveal different weaknesses.

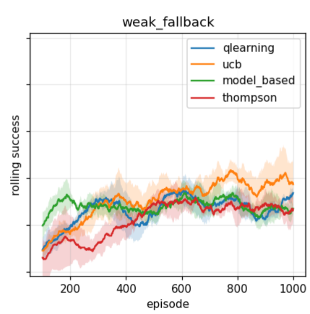


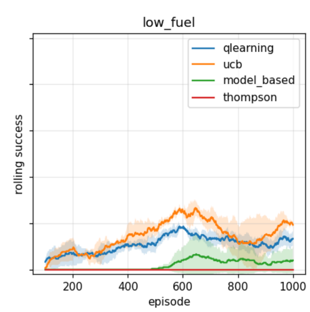
(a) default





(b) risky *(c)* harsh hazard





(d) weak fallback *(e)* low fuel

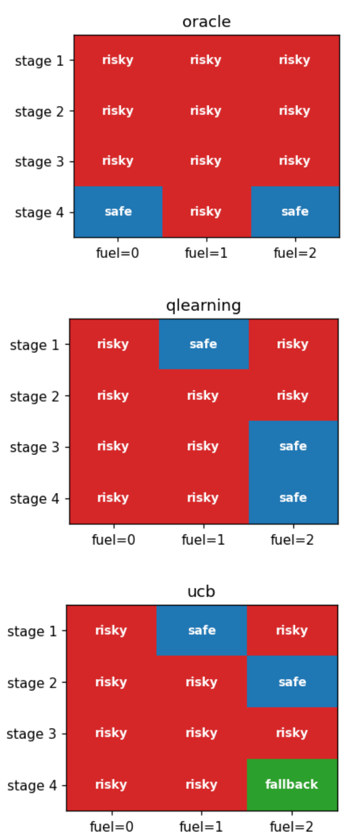
Figure 4. Success rate under default conditions and four environment variants

Increasing the hazard penalty makes all agents more conservative and lowers success. Reducing initial fuel is the hardest setting because it removes most of the strategic benefit of preserving fuel for later recovery. Doubling risky failure probability pushes all agents toward safer strategies and especially hurts methods that rely more heavily on optimistic exploration. Weakening fallback has comparatively little effect because fallback is not used often in the best learned policies.

Across these variants, UCB and Q learning are the most robust on average. This is an interesting nuance: the agent with the best default environment performance is also the one that remains competitive across substantial task perturbations.

4.5. Learned policies versus the oracle

Figure 5 provides the strongest qualitative check on the learning outcome. Despite quantitative differences, the learned policies broadly recover the same strategic pattern as the oracle: risky in early stages when fuel preservation has high long term value, and safe in the final stage when enough fuel remains to lock in success. This agreement is important because it shows that the benchmark is not merely distinguishing agents by noise or luck; it is distinguishing them by how efficiently they discover the correct fuel and risk trade off.



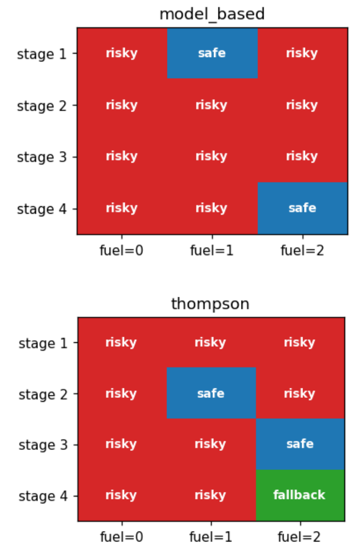


Figure 5. Greedy policy heatmaps at (hazard = 0, fallback = available)

5. Discussion, Reproducibility, and Limitations

The project satisfies the main hypothesis in a nuanced way. Uncertainty aware methods do improve over plain Q learning, but not all such methods do so equally. UCB is the best overall method in this environment. Model based value iteration is also strong once its transition estimates become accurate. PSRL exhibits an interesting safety and performance trade off: it has lower hazard exposure, but lower success and higher regret than UCB and Q learning.

Several caveats matter for interpretation. First, the oracle success ceiling is only 47.8%, so no algorithm can achieve near perfect performance here. Second, five seeds are enough to show stable trends, but not enough for strong significance claims. Third, the single episode trajectory traces are useful for explanation but should not be over interpreted. Fourth, the environment is intentionally small and tabular; these results should therefore be read as a controlled study of exploration behavior rather than as a realistic spacecraft controller.

For reproducibility, the submitted implementation includes a main notebook, a source tree with environment and agent modules, unit tests for the environment, and figure assets exported from the notebook. The project can be reproduced by installing the package, opening the main notebook, and running all cells top to bottom. All reported experiments use the fixed protocol described in Table 3; the five evaluation seeds are 0, 1, 2, 3, and 4, and the hyperparameters were fixed before the final benchmark rather than tuned separately for each agent or variant. We report mean and standard deviation across seeds; formal bootstrap confidence intervals are left for future work because the project scope focused on implementation, comparison, and qualitative policy analysis.

6. Conclusion

We presented a compact risk aware lunar mission benchmark that makes the trade off between fuel preservation and catastrophic risk explicit. Because the environment is small enough for exact planning, it provides both a meaningful RL application and a controlled comparison setting with an oracle upper bound. Among the four tabular methods studied here, UCB achieved the best overall performance, while model based value iteration became competitive after a short data collection phase. PSRL showed the lowest hazard rate but did not dominate on success or regret, so its behavior should be understood as a safety and performance trade off rather than an overall win. The results support a nuanced version of the main claim of the project: exploration strategies that reason more explicitly about uncertainty can outperform standard Q learning, but only when the uncertainty mechanism helps the agent discover useful actions without becoming overly conservative.

A. Additional Experimental Results

The plots of figure 6 show final success rate, final mean return, hazard rate and cumulative regret for a standard mission. It supports the same conclusion as the main result table 4.

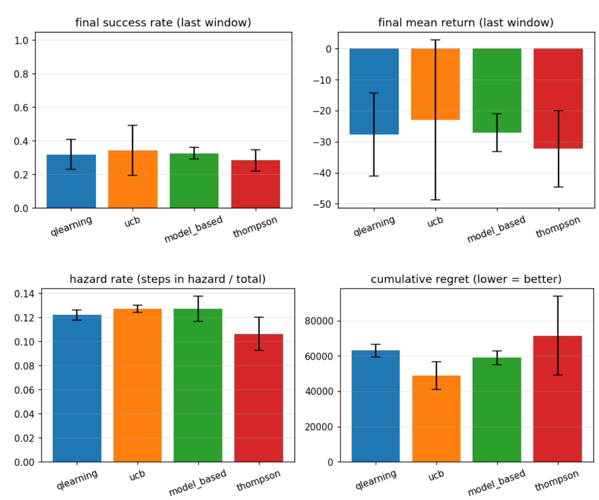


Figure 6. Additional summary of the final benchmark metrics

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LLM Use Disclosure

The main project design was developed by the authors through group discussions. This includes the problem formulation, lunar mission setting, MDP formulation, state and action spaces, transition logic, reward structure, choice of algorithms, experimental protocol, interpretation of results, and final conclusions.

Cursor AI was used mainly for the coding related parts behind Sections 2, 3, and 4, including assistance with the environment implementation, agent modules, experiment scripts, plotting utilities, notebook structure, debugging, and code organization. OpenAI ChatGPT was used throughout the report to support writing, improve clarity and structure, refine the explanation of the reinforcement learning formulation, check alignment with the grading rubric, and help identify suitable evaluation metrics such as success rate, return, hazard rate, and regret. All LLM suggestions were reviewed, adapted, and validated by the authors, who remain responsible for the correctness of the implementation, experiments, analysis, and submitted report.